

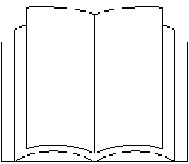


Operator's manual for Attachment



Backhoe 170

Product number: 300-700-series loaders: A33748



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or damage to property:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or damage to property. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This operator's manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an user with little experience can use the backhoe and the loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the equipment have received proper guidance and familiarised themselves with the manual and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

The AVANT backhoe 170 is an attachment suitable for use with 200-series AVANT multi purpose loaders. It is equipped with a hydraulic control valve unit to control its movements, and it can be slewed for efficient operation. The backhoe is well suited for digging and loading of soil to a trailer even at narrow areas. The backhoe features adjustable boom speed for easier operation. As standard, the backhoe is equipped with a 250 mm bucket, and also a 400 mm bucket is available as an option. The bucket can be changed quickly thanks to the pin locking of it.

The use of the backhoe as a lifting crane is prohibited due to a risk of tipping the loader-backhoe combination over. The backhoe is not suitable for dredging.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Not all maintenance tasks can be performed by the operator; more demanding service or repairing tasks must be left to qualified professionals. Appropriate protective equipment and original spare parts must be used for the maintenance. Read the maintenance and servicing instructions in this manual. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for the backhoes

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not familiarised yourself with its operation and the related hazards.

The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- monitor and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- When mounting the attachment to the loader, ensure that the locking pin of the loader's quick attach plate is in the lower position and that it has **locked the attachment to the loader**. Always mount also the supporting bracket. Carefully read the instructions of how to attach the backhoe to the loader.
- Keep the control valve boom of the backhoe fastened during mounting or taking the backhoe off from the loader to prevent it from tipping over.
- Start the use of the backhoe carefully and calmly.
- Keep others away from the operating area of the backhoe due to crushing hazard.



- Do not use the backhoe on horizontally tilted terrain. Ensure the stability of the backhoe and the load carrying capacity of the ground even at level terrains.
- When working, pay attention to the contours of the terrain and other hazards, such as rocks or branches and trees reaching the driver's seat. Pay attention to the surroundings and any other persons and machines moving in the vicinity.
- Always transport the backhoe as low and as close to the machine as possible to keep the centre of gravity low. For transporting, turn the boom of the backhoe to the position shown in section 6.
- Turn the backhoe to a position in which it will be stable during storage.
- Shut down the loader engine before any cleaning, maintenance or adjustments. Ensure that the backhoe is supported properly supported when going near the attachment.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual.



Note that both the boom of the loader and the backhoe move down even if the engine has been stopped. Never go under a raised boom or the attachment. Lower the backhoe on the ground before leaving the driver's seat.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.

Remember to wear proper protective clothing:



- The noise level at the driver's seat exceeds 85 dB(A). Wear hearing protection while working with the loader.

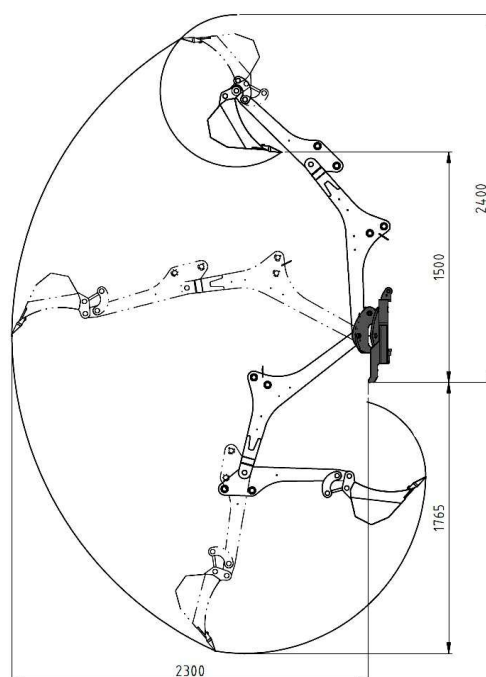


- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.

4. Backhoe specifications

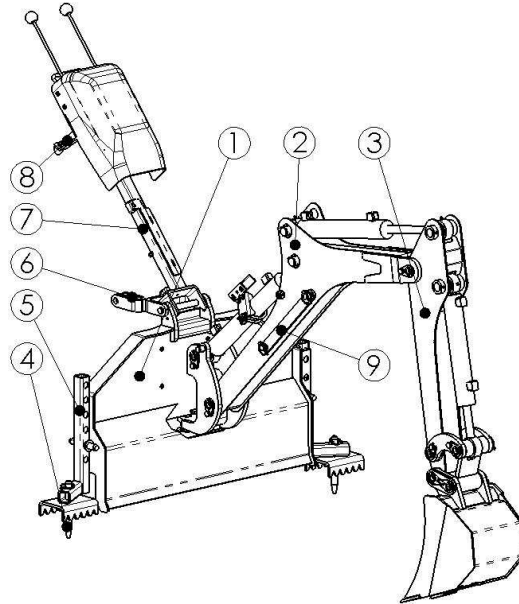
Table 1 – Backhoe 170 technical data

Product number	A33748
Digging depth:	1 700 mm
Loading height:	1 500 mm
Weight:	180 kg
Standard bucket:	250 mm, 2 teeth
Bucket options:	400 mm A33960, 3 teeth
Suitable Avant loader:	216, 218, 220



Main components of backhoe:

1. Frame
2. Lifting boom
3. Front boom
4. Outrigger tire
5. Adjustable outrigger
6. Supporting bracket (the top link)
7. Valve boom
8. Valve boom bracket
9. Service support



Listed below are the markings and labels fastened to the backhoe, and they must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer or contact information provided on the cover page.



The warning decals contain important safety information. Replace damaged or missing decals with new ones.

CE	AVANT TECNO OY Ylötie 1 FIN-33470 YLÖJÄRVI		
TYPE	Axxxxx	No.	00000
Year	2010	kW	kg

- A46771 – Read the instructions before use.
- A46772 – Do not go under a raised attachment; keep a safe distance from raised or freestanding equipment.
- A46803 – Crushing hazard, keep clear of moving parts.
- A46797 – Crushing hazard in the reach area of the backhoe, keep bystanders away from danger area.

Identification plate of the attachment



A46771



A46772



A46803



A46797

5. Assembly of the backhoe

The backhoe 170 is ready for use as delivered. To mount the backhoe, in addition to the quick coupling and connecting the hydraulic fittings the supporting bracket (also called the top link) must also be mounted. To make the coupling of the backhoe 170 tight, on the front side of the backhoe frame there are also four screws for reducing the slack between the attachment and the loader quick coupling plate.

Mounting the attachment to the loader is quick and easy, but it must be done carefully. The backhoe may tip over on the loader or the operator, if it has not been properly attached. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3.



Beware of the hinged control valve boom of the backhoe. Do not unfasten the valve boom strap until the backhoe is completely mounted to the loader.

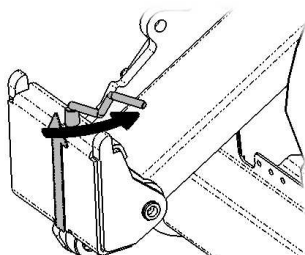
5.1. Mounting the backhoe to the loader

The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. In addition to this, a locking bracket must be also installed.



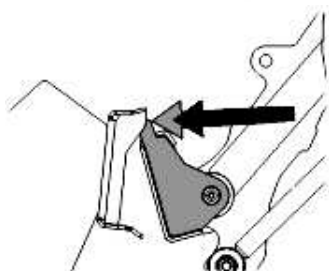
Never move or raise an attachment that has not been locked.

The backhoe is mounted to the loader as follows:



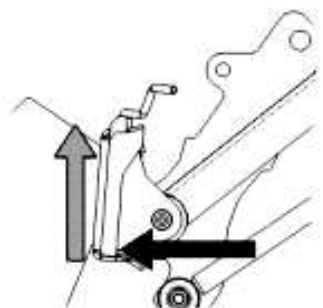
Step 1:

- Lift the quick attach plate locking pin up and turn it backwards into the slot so that it is locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



Step 2:

- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment.



Step 3:

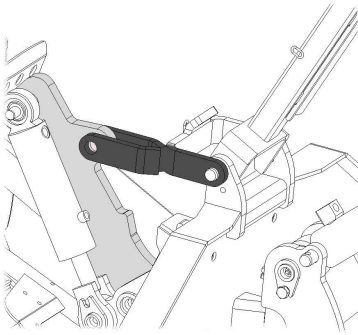
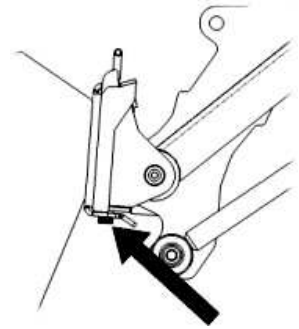
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pin manually.



Do not use the automatic locking of the locking pin with the backhoe. The centre of gravity of the backhoe is at relatively high, which creates a risk of the backhoe tipping over.



Before moving or lifting the attachment, ensure that the locking pin is in the lower position and comes through the fastener.

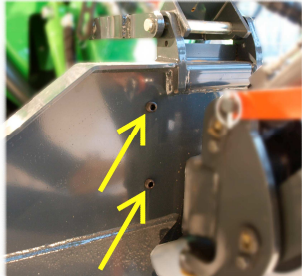


Step 4: Fitting the support bracket

- Lift the backhoe slightly off the ground. Remove the pin from the loader side of the supporting bracket and place the bracket on the bracket of the loader boom.
- Tilt the backhoe to align the holes.
- Fit the locking pin through the holes and lock it with the cotter pin.



Do not use your fingers to check the alignment of the support bracket. A small movement of the backhoe could pinch or shear a finger.



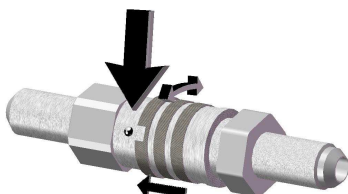
Step 5: Tightening the quick coupling plate

- To remove the slack between backhoe and the loader quick attach plate, use an 8 mm allen key to tighten the four screws on front of the backhoe frame.
- Tightening is recommended when using the backhoe for longer periods.

5.2. Connecting and disconnecting the hydraulic quick couplings

The hydraulic couplers of the backhoe should be connected so that the male coupling is connected to the coupling marked with a coloured cap on the loader. The other hose is connected to the tank line (free return line). Before connecting or disconnecting hydraulic couplings, shut down the loader engine and move the auxiliary hydraulics pedal on the loader to its extreme positions a few times. This way, there will be no remaining pressure in the system and the fittings are easy to connect.

The hoses can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn unintentionally into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause injuries.

Keep the hydraulic components as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

If residual pressure is left in the system, it is often possible to disconnect the fittings, but connecting them the next time may be difficult. If the hydraulic fittings will not connect due to the pressure in the system, the pressure must be relieved from the system. First try it by moving the control levers of the backhoe and note that the backhoe may move a little when doing so. If this doesn't help, lifting the backhoe boom a little or pressing its bucket against the ground with the loader may be helpful.

In other cases the pressure left in the system must be relieved by opening the fitting slightly with tools. Keep both of the hoses disconnected and wrap the fitting in a cloth to prevent oil spillage. When the fitting is opened slightly, a small amount of oil will be released; beware of the oil. Tighten the fitting immediately after the procedure.



The backhoe may move when the pressure is released, stay clear from the equipment. Wear protective goggles and gloves, and wrap the fitting with cloth when releasing pressure from the hoses to prevent oil spilling.

5.3. The valve boom of the backhoe

Under the valve boom there is a supporting bracket, which should lean against a suitable place of the loader boom. Note that the controls valve are at their operating position only after the backhoe is set on its supporting stands as shown in section 6.1.1. If you want to have the control valve closer to you, the boom can be adjusted with the adjusting screw on the side of the boom.



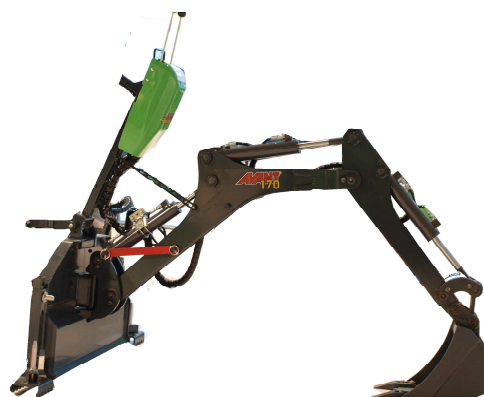
Note that tilting the boom may not be possible with extended boom, as it would hit the safety frame of the loader. Do not allow the valve boom to tip over; handle it with secure grip with both hands.

5.4. Taking the backhoe off from the loader



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Refit the protective caps on the fittings to prevent impurities from entering the hydraulic system.

When finished working with the backhoe and setting it for storage, first turn it's slewing to the middle position. Tilt and lock the valve boom on the backhoe. Lower the backhoe on level ground on its outriggers and turn the backhoe to the position showed in the adjacent picture for best stability.



Storage position of the backhoe

Tilt the loader front plate so that it possible to remove the supporting bracket. After removing the bracket tilt the backhoe a little forward to a position in which it will stay stable on its own. Shut down the loader and lower the bucket onto ground. Move the control levers of the backhoe to all directions, as well as the auxiliary hydraulics control pedal of the loader to their extreme positions to release any residual pressure, and disconnect the hydraulic couplings.

6. Instructions for use

Work can be started when the backhoe has been mounted and the supporting bracket and the hydraulic couplings have been connected according to the instructions. Check the attachment once more before starting work.

- Check that all obstacles, including any hidden ones, have been removed from the working area before operation. Remember correct working methods and avoid unnecessary descending from the driver's seat.
- Make sure that it safe to dig in the location. Find out if there are electric cables, water lines or similar obstacles at a depth that can be reached with the digger.
- Operate the backhoe only at well-lit areas.
- Ensure that the loader and attachment are used in a safe manner and as intended.
- Check regularly to ensure there are no hydraulic leaks. Refer to Section 7 for maintenance instructions.
- Operate the attachment from the driver's seat only.

6.1. Operation

To operate the backhoe, lock the auxiliary hydraulics control pedal of the loader on. When operating with the backhoe, moderate engine rpm is often enough. Higher rpm will not increase the digging force, but if the engine stalls, use a little higher revolutions. Keeping the engine rpm low according to operating conditions will save fuel.

Keep the auxiliary hydraulics turned off when transporting the equipment or whenever not using the backhoe. Turn the boom slewing to its middle position so that it will not create a force that could affect the stability of the loader.

At the work site check the operating environment and set the backhoe on its outriggers, as shown in section 6.1.1. On sloped terrains avoid pulling material from the sides, which could affect the loaders stability. Preferably drive the loader so that digging is performed from the front of the loader.

Check the operating area before starting to work:



- Ensure that there are no other persons at the work site. Do not allow others closer than five meters from the backhoe, near the danger area of the boom or directly in front of the backhoe.
- Make sure that it is safe to dig at the site. Find out if there are any electric cables, water lines or similar obstacles within the reach of the equipment.



Turn the auxiliary hydraulics of the loader off whenever not operating the backhoe or when transporting the backhoe. The control levers of the backhoe may be accidentally operated, so that accidental movement of the backhoe may create a hazardous situation.



Do not use the backhoe without the supporting bracket. The purpose of the bracket is to lock the tilting and to make the backhoe more stable for more effective operation.



On inclined terrains drive carefully. Do not operate the backhoe at horizontally tilted terrains. Transport the backhoe as low and close to the loader as possible.

6.1.1. Using the outriggers

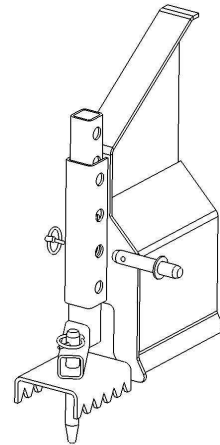
The backhoe is equipped with outriggers, which are intended to stabilize the backhoe during operation and when moving material to the sides. Using the backhoe is more efficient with the use of the outriggers, so it is advisable to set them correctly.

In delivery configuration the outriggers are placed upside down. The outriggers are mounted so that their feet point to forward direction. Note the markings on the outriggers; the left and the right ones are dissimilar.

The height of the outriggers can be adjusted for operation on uneven terrains. There are several holes on the backhoe frame and the outrigger. Use the cotter pin to lock the adjustment pin.



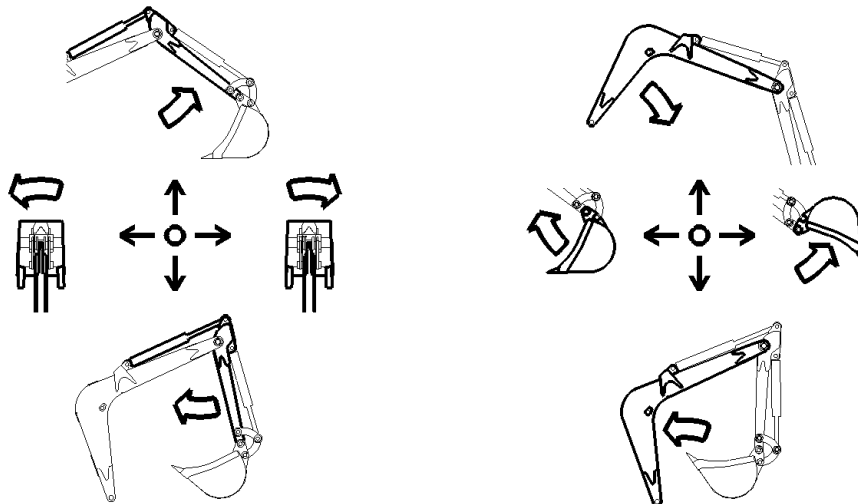
Adjust the outriggers so that the front tires of the loader will rise slightly of the ground when the loader boom is set to its lowest position. This way the control levers will be easy to reach and the best stability can be achieved.



The outriggers are equipped with spikes, which can be used to keep the backhoe better in place during digging in soft ground. The spikes should sink into the ground for a better grip from the ground. However, when operating on hard surfaces, the spikes can be turned facing upward, making the outriggers function as usual. The spikes are attached to the outriggers with ring splints, which makes it easy to use them.

6.1.2. Digging with the backhoe

The backhoe is controlled with its two control levers. The backhoe has relatively many simultaneous movements, to which it is advisable to become familiar with time and away from other persons or buildings. The operation of the control levers is shown in the figure below and on the next page. See also instructions to adjust the speed of the movements of the backhoe from section 6.3.



Backhoe controls, left and right control lever

With the left control lever the front boom of the backhoe and the slewing are controlled. The right lever controls the main boom and the bucket of the backhoe.

Digging with the backhoe is the most efficient when the edge of the bucket is kept at an angle that allows the edge to cut the soil the most effectively. Pull the edge of the bucket towards the loader with the left lever and tilt the bucket by operating the right control lever at the same time. The bucket should not be pressed too hard downwards, since this would result in lifting the front tires of the loader, and digging becomes ineffective.



Beware of the caving in hazard in the excavation or near it. Do not drive near the edge of the hole or allow children to play near it.

At the first use the movements of the backhoe may be imprecise as there might be air trapped in the hydraulic components. The hydraulic system will bleed itself, when moving all the movements to their extreme positions one by one.

6.2. Adjusting the speed of the backhoe movements

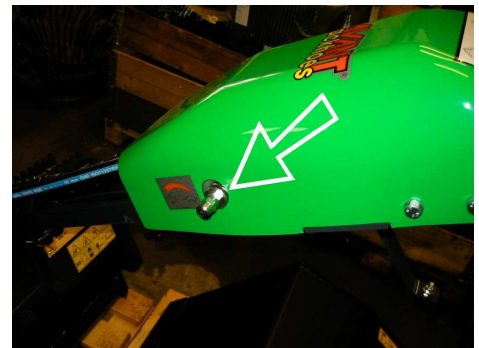
The speed of the backhoe movements can be adjusted with the flow divider valve, the adjusting screw of which is visible at the left side of the control valve cover. When learning to use the backhoe the actual working speed can actually be greater due to increased accuracy, even if the movements of the backhoe are slower.



Note that adjusting the speed of the movements affects only the speed, not the force of the backhoe.

The speed of backhoe movements can be adjusted as follows:

1. Shut down the loader and release the residual pressure as described in section 5.2.
2. Open the cap of the adjusting screw with a 14 mm spanner
3. Loosen the locking screw of the adjusting screw with a 14 mm spanner
4. Turn the adjusting screw with an Allen key desired direction.
5. When a suitable setting has been made, tighten the locking screw and fit the cap.



6.3. Changing the bucket

The bucket of the backhoe 170 is mounted with two pivot pins. Change the bucket when the backhoe is mounted to the loader and use the service support described in section 7.1. Place the bucket near the ground. Remove the cotter pins and support the bucket while removing it. Use a soft hammer or a block of wood to soften the hits, so that the pins will not get damaged. Do not hit directly at the pins. In most cases it is easier to remove the front pin first. When mounting the new bucket remember to lock the cotter pins in place.

7. Maintenance and servicing

Continuous maintenance of the backhoe includes regular cleaning and lubrication. The condition of the hydraulic hoses and components should be monitored. Due to crushing hazard caused by movements of the backhoe booms, all maintenance work must be done when the backhoe has been lowered down completely and any additional required supports are used.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a fault in the hydraulic system. Leaking hydraulic fluid may penetrate the skin and cause serious injuries. Hydraulic fluid is also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of the hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component in question must be replaced. Spare parts are available from your nearest AVANT retailer or service point. Leave the service of hydraulic components to a trained professional if unsure how to safely perform the task.

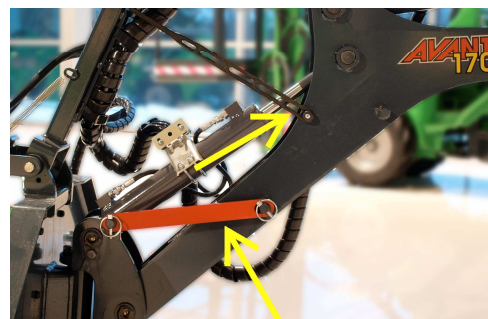


Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a fault in the hydraulic system.



7.1. Service support

At the side of the backhoe lifting boom there is a red service support, which is intended to lock the booms for transport, storage and service procedures. The service support is fitted so that the other end of it is mounted so that its other end is locked to the bracket near the slewing pivot. During maintenance the backhoe will be the most stable when it is mounted on the loader and its outriggers are against the ground.



All maintenance procedures must be performed when the bucket has been lowered down completely. The backhoe and its bucket may lower without operating the control levers and cause serious injuries.

7.2. Cleaning the equipment

You may use a pressure washer and a mild detergent. Do not use strong solvents or spray directly at the hydraulic components or at the labels on the attachment either. Lubricate the lubrication points of the attachment after washing, shown in section 7.3

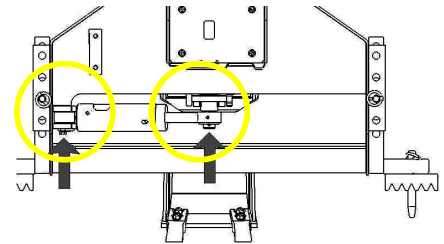
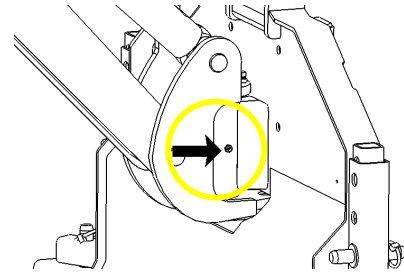
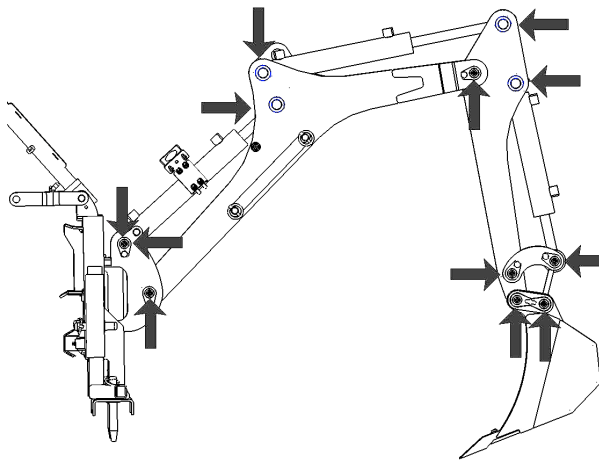
Do not let water to stay in the bucket. For longer storage periods touch up the paint where needed, grease the lubrication points and oil the visible parts of hydraulic cylinder shafts to prevent rust damage. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example.

7.3. Lubricating the equipment

There are a total 15 lubricating points on the backhoe; they are marked in the figures below. Lubricant should be added every 5-10 hours of use. In the pivot pins and the cylinders the grease nipples are standard R1/8" nipples, replace any damaged nipple.



Lubricant must be added approximately every 5 to 10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.



8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) for the attachment from the date of purchase.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work, such as the replacement of blades or the belts, or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveysvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

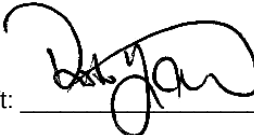
We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:
 SFS-EN ISO 12100-1, SFS-EN ISO 12100-2, SFS-EN 982, SFS-EN 474-4

Kuvaus: Avant-kuormainten kanssa käytettävä kaivuulaite
 Beskrivning: Grävaggregat; arbetsredskap för Avant lastare
 Description: Backhoe bucket; attachment for Avant loaders
 Beschreibung: Anbaubagger; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant Mallit / Märken / Models / Modellen:	200 series
170	A33748

Allekirjoitus / Namnteckning / Signature / Unterschrift:  10.6.2010 Ylöjärvi, Finland

Risto Käkelä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]